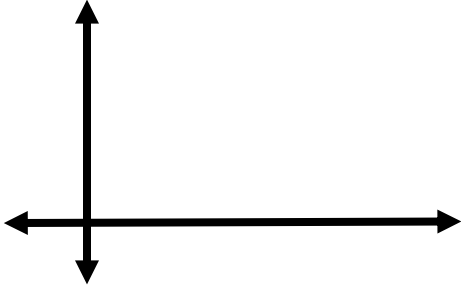


The Second Derivative Test for Concavity

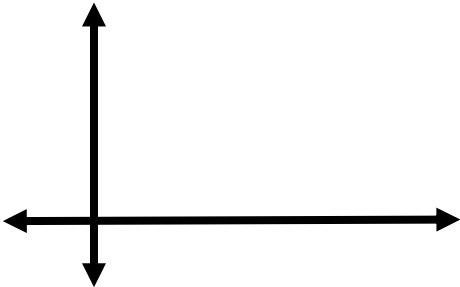
Concavity

- Concave Up (C. U.)



- bent so that the graph opens upward
- $f''(x)$ is positive over this interval
- tangents are below the graph

- Concave Down (C. D.)

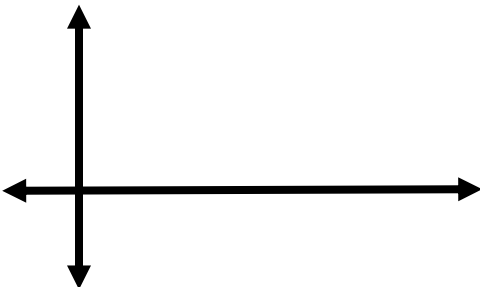


- bent so that the graph opens downward
- $f''(x)$ is negative over this interval
- tangents are above the graph

Point of Inflection (P. O. I.)

The point where the graph changes from concave up to concave down or vice versa is called a P. O. I.

Graphically,



Note: $f''(x) = 0$ at a P. O. I.

Test for Concavity

- If $f''(x) > 0$ for all x on some interval, then the graph of $f(x)$ is concave up on that interval.
- If $f''(x) < 0$ for all x on some interval, then the graph of $f(x)$ is concave down on that interval.

To Find Possible P. O. I.

Solve $f''(x) = 0$ or determine where $f''(x)$ DNE. Use an interval chart to determine on which intervals $f''(x)$ is positive or negative.

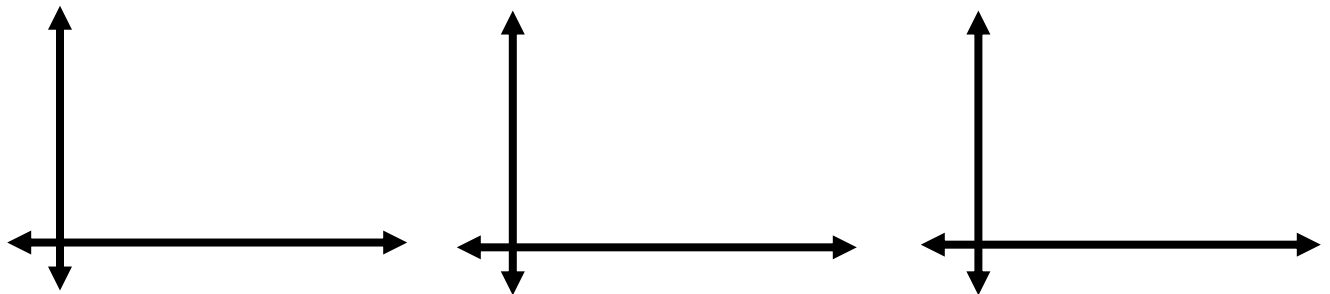
Example 1: Determine where the following curves are C. U. or C. D. and locate any P. O. I.

a) $f(x) = x^3 - 3x^2 + 4x$

b) $f(x) = \sqrt[3]{x}$

Cusps and Vertical Tangents

- A cusp is a pointy bend on the graph where no unique tangent line exists.
- A cusp is a good example of the situation where a function can have a maximum or a minimum at a point where the derivative of the function DNE.
- To locate a cusp, determine where $f'(x)$ DNE and then test the concavity on either side of that critical number.
 - If the concavity remains the same there is a **cusp**.
 - If the concavity changes there is a **vertical tangent**.



Example 2: Verify that $f(x) = x^{\frac{2}{3}}$ has a cusp at $x = 0$.

Test for Local Extrema using the 2nd Derivative Test

Given c is a critical number, then

→ If $f'(c) = 0$ and $f''(c) > 0$, then $f(x)$ has a local minimum at c .

→ If $f'(c) = 0$ and $f''(c) < 0$, then $f(x)$ has a local maximum at c .

→ If $f'(c) = 0$ and $f''(c) = 0$, then the 2nd Derivative Test fails.

Example 3: Use the 2nd Derivative Test to find the local extrema of $f(x) = 3x^5 - 5x^3$.